

Experimental Restoration of Disturbed Cliff-Edge Forests in Bruce Peninsula National Park, Ontario, Canada

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Abstract

Restoration of degraded natural vegetation in parks is often complicated by the need to maintain public access. We tested whether the natural canopy species, *Thuja occidentalis*, can be restored to degraded cliff edges in Bruce Peninsula National Park, Ontario, Canada without reductions in visitor numbers. Eighty 10-year-old and 80 4-year-old container-grown saplings and 1,400 seeds were planted and monitored for 4 years. Eight treatments were applied that tested for effects of planting site (distance from cliff edge and pathways) and supportive measures (soil, water, cages, or signs) on survival, growth, and damage. No trees became successfully established from seed. Younger trees showed faster initial establishment and growth, but 4-year survival was the same for both age groups (39%). Supplemental soil improved the health of planted trees, and both soil and water slightly improved their survival. Cages did not affect survival and growth but decreased damage to

4-year-old trees and increased it for 10-year-olds. Signs had no effect on any measured variable. Trees planted away from the cliff edge and from pathways had the greatest establishment success, 4-year survival, and general health. Relative visitor density, as estimated from spot counts of visitors, had the largest effect on restoration success; the results suggest a threshold of visitor density above which restoration may be impossible. Planting location, especially with respect to shade, was also important. The planting of 4-year-old trees without supportive measures is suggested as the most cost-effective restoration technique at this site. The results indicate that restoration of open woodland habitats is possible without total visitor exclusion but that some restrictions on visitor numbers or activities are necessary.

Key words: disturbance, trampling, *Thuja occidentalis*, canopy effects, survival, damage.

Introduction

In parks and conservation areas with large numbers of visitors, managers often face the dilemma that the natural environment has become severely degraded, but closure of popular tourist destinations for restoration purposes reduces revenues from visitors. An alternate course of action is to attempt restoration in the presence of continued disturbance, but the effectiveness of such an approach has never been demonstrated. A large-scale restoration program would be ill advised until a small-scale program of experimental restoration is completed (Fimbel & Kuser 1993; Walmsley & Davy 1997a, 1997b; Stange & Shea 1998; Zink & Allen 1998).

In the restoration of forested ecosystems, the creation of a canopy is the most critical issue because of its influence on the microclimate of the whole community. For this reason, one approach is to restore the canopy trees first and either add or encourage natural invasion by understory plants later as shade develops (Harker et al. 1993). Seeds are the most economical technique for restoring large sites and have been used successfully for trees,

but transplants or potted nursery stock are often preferred because their establishment time is shorter (Harker et al. 1993). Regardless of the material used for restoration, the preparation of the site before planting and the care given to the plantings during the first year are thought to be closely related to their success. For example, maintaining adequate soil moisture during the establishment period is thought to be important, although irrigation is not considered necessary after the first year (Howell 1986; Harker et al. 1993).

In Bruce Peninsula National Park, Ontario, hiking trails along the cliff edges of the Niagara Escarpment offer a spectacular view of the waters of Georgian Bay and may be used by more than 20,000 visitors in a season (Kettle 1998) (Fig. 1). In undisturbed parts of the park, the cliff edges support a characteristic stunted old-growth forest community, which is dominated by the woody species *Thuja occidentalis* (eastern white cedar) (Larson et al. 1989). Heavily visited areas along the cliffs show a visible degradation of the cliff-edge forest, and the most disturbed sections of cliff edge are virtually denuded of vegetation (Larson 1990; Parikesit et al. 1995). Concern about the carrying capacity of the shoreline trails has led park staff to consider imposing limits on admissions, although complete closure is not being considered. The present study was undertaken to explore the possibility of restor-

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Figure 1. Indian Head Cove within Bruce Peninsula National Park, Ontario, at the height of the summer visitor season (August). The shoreline shown in the foreground is part of block 7; the rise in the background forms block 8.

ing the canopy species, *T. occidentalis*, to degraded cliff edges without reducing visitor numbers. We asked the following questions. What is the likelihood that such a restoration would be successful within the range of current disturbance levels? What are the best protocols to use in a future large-scale restoration of this site? Specifically, we investigated the effects of planting seed or differently aged nursery stock, choosing different planting locations with respect to distance from the cliff edge and pathways, attempting to improve growth conditions for plantings by adding soil and watering, and attempting to protect plantings from human disturbance using cages and signs.

Methods

Study Area and Experimental Design

The study was conducted along a 360 m long and 10 m wide section of Niagara Escarpment cliff edge that included two of the major points of interest within Bruce

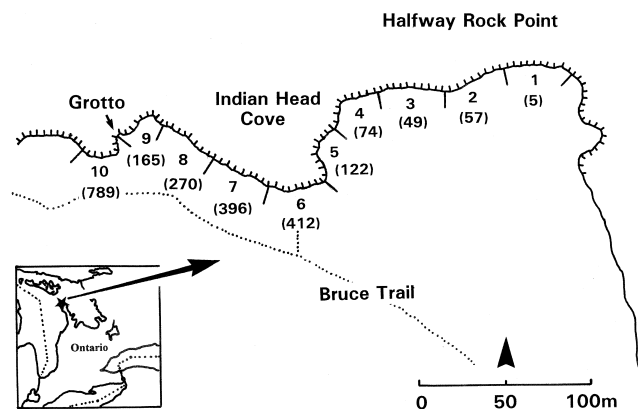


Figure 2. Map of the study area, showing the location of the 10 blocks and the number of visitors counted in each block as a measure of relative visitor density.

Peninsula National Park, the Grotto and Indian Head Cove, as well as adjacent less frequently visited parts of the shoreline (Figs. 1 and 2). To stratify for differences in current disturbance levels, the sampling area was divided into 10 blocks of 36 m each and the relative disturbance within each was quantified by conducting visitor counts. Over a 2-hr period on three separate dates (21 July, 5 August, and 1 September 1999), simultaneous counts were taken every 5 minutes of the number of people present within each block. All observation periods were then summed to provide a relative measure of visitation rate for each block.

Each block was divided into 24 sections of 1.5 m length, and these were randomly assigned to three tree age classes and eight treatments. A planting site, consisting of a soil-filled depression or crevice in the rock, was selected within each section. Such planting sites reflect the natural habitats of the trees in this ancient forest (Matthes-Sears & Larson 1995). Sites were located either within a meter of the cliff edge or 10 m away from the edge, depending on the treatment (Table 1). At the 10-m distance, where visitor traffic has created a dense network of small footpaths, planting sites selected were either disturbed (on pathways) or undisturbed. No such distinction was possible for cliff edge sites where disturbance was uniformly heavy. Each planting site was further categorized by canopy cover as either shady, open, or intermediate.

Plant Material

Three age classes of *T. occidentalis* were used for planting: 10-year-old trees, 4-year-old trees, and seeds. All seed had been collected in the autumn of 1996 from cliffs in southern Ontario and was stored at 4°C. Some of the seed was pregerminated in a growth chamber at 30°C/8-hr light and 20°C/16-hr dark on moist filter paper and was used when the radicle was starting to protrude. The trees had been grown from seed and kept in pots outdoors at the University of Guelph.

Table 1. Treatments applied to planted trees and seeds and treatments compared when testing specific hypotheses using contrasts.

<i>Treatment No. and Name</i>	<i>Description</i>
1, edge – soil	Planted at cliff edge, no supplements
2, edge + soil	Planted at cliff edge with added soil
3, edge – soil + water	Planted at cliff edge without added soil, watered weekly during the first growing season
4, edge + soil + water	Planted at cliff edge with added soil, watered weekly during the first growing season
5, edge + soil + cage	Planted at cliff edge with added soil and protective enclosure
6, edge + soil + sign	Planted at cliff edge with added soil and sign
7, 10 m in, on path	Planted 10 m back from edge, on a pathway, with added soil
8, 10 m in, not on path	Planted 10 m back from edge, in undisturbed location, with added soil
<i>Effects Tested For</i>	<i>Treatments Compared</i>
Improved physical growth conditions	2, 3, 4 vs. 1
Added soil	1, 3 vs. 2, 4
Added water	1, 2 vs. 3, 4
Efforts to minimize damage by humans	5, 6 vs. 2
Protective enclosures	5 vs. 2
Signs	6 vs. 2
Distance from cliff edge	7 vs. 2
Trampling	8 vs. 7

Experimental Treatments

The eight treatments (Table 1) were designed to test the effects of ameliorating growth conditions by adding soil, water, or soil and water; protecting trees with brightly colored signs or cages; and varying the planting location with respect to disturbance and distance from the cliff edge. Soil addition took place at planting and consisted of 500 mL of commercial potting mix (Pro-Mix; Premier Horticulture, Rivière-du-Loup, Québec) for 4-year-old trees and seeds and twice that amount for 10-year-old trees. Watering was performed during the first growing season only and consisted of a once weekly soaking of the soil surrounding the tree or seeds. Cages were open at the top and were constructed of poultry netting 80 cm high for 10-year-old trees, 60 cm high for 4-year-old trees, and 40 cm high for seeds. Signs were of yellow plastic, 15 cm in diameter, and inscribed “Replanted tree—please do not disturb.” They were placed immediately adjacent to the planting site.

Planting

Each planting location received one tree or several seeds. The trees were planted during the last week of May 1997.

Five pregerminated seeds per site were planted during the first week of June 1997. When few seedlings emerged, another five germinated seeds were added to the same sites in half of the blocks 2 weeks later. A third planting of 10 ungerminated seeds per site was performed in April 1998. All trees and seeds were watered once immediately after planting.

Measurements

Each planting site was visited for data collection at the following intervals: weekly, biweekly, and monthly in the first, second and third growing seasons, respectively, and at the beginning (May) and end (September) of the fourth growing season.

Damage and shoot growth were assessed on five (10-year-old trees) or three (4-year-old trees) randomly selected branches plus the main leader of each tree. New branches were selected and tagged at the beginning of each growing season. At each visit, a score between 0 and 5 was assigned to each branch in three categories: breakage, representing damage caused by humans (most often by trampling); herbivore damage; and browning (also including yellowing or wilting) as a sign of general poor health of the tree. To measure shoot growth, a small mark of nontoxic paint was placed on the leader of the branch and the distance from the mark to the growing tip was measured (Matthes-Sears et al. 1995). Also at each visit, the angle of the main axis with respect to the horizontal plane and the presence of exposed roots were scored on a scale of 1–3.

At the end of the study, each tree was assigned to one or more fate categories (Table 2) that took into account not only whether the tree survived but also its health, exposure to disturbances, and likely reason for mortality.

Analysis

Statistical analyses were performed using SAS (v. 6.12; SAS Institute 1989). Life table analysis (Fox 1993) was conducted to compare survivorship curves among different treatments, blocks, and tree ages. PROC LIFETEST was used to generate Wilcoxon chi-squares for overall tests. Z-statistics for multiple comparisons were calculated as outlined by Fox (1993). The distribution of tree fates among treatments, blocks, and age classes as well as between shady versus open site categories were analyzed using Fisher's exact test of independence.

Damage scores were averaged over all branches of each tree and graphed against measurement date to reveal seasonal trends. For the data collected at the beginning and end of each growing season, analyses of variance (ANOVAs) were performed to test for effects of block, age, treatment, and age $\times$ treatment on damage scores. Contrasts were used to reveal the effects of specific treatments (Table 1). Logarithmic, arcsine square root, or inverse transformations were used on the data where necessary to satisfy the normality assumption of ANOVA.

Table 2. Tree fate categories, the number and percentage (out of 160) of trees falling in each category, and description of the category.

Fate Category	No. of Trees	Percentage of Trees	Description
Trees alive at the end of the study			
1	33	21	Healthy throughout: Trees that were healthy throughout the study period
2	33	21	Variable health:
a	9	6	Trees that experienced both periods of health and decline and were unhealthy at the end of the study period
b	24	15	Trees that experienced both periods of health and decline and were healthy at the end of the study period
3	20	12	Survived partial uprooting: Trees with roots that were partially or temporarily exposed during the study but that survived
4	11	7	Survived trampling: Trees that were severely or repeatedly trampled during the study but survived
5	7	4	Survived herbivory: Trees that were severely or repeatedly chewed but survived
6	6	4	Survived cage damage: Trees that were damaged by the cage but survived
Trees that died during the study period			
7	24	15	Failed to become established: Trees that never took root and died soon after planting, without external factors contributing to their death
8	14	9	Died without external factors: Trees that experienced both periods of health and decline but then gradually declined and died without external factors contributing to their death
9	39	24	Uprooted:
a	10	6	Trees that were healthy, then suddenly disappeared or were completely uprooted and died
b	16	10	Trees that were unhealthy and might have died for other reasons, but before then suddenly disappeared or were completely uprooted and died
c	13	8	Trees that were healthy until partial or temporary uprooting caused them to slowly decline and eventually die
10	29	18	Trampled:
a	11	7	Trees that died from trampling alone: were healthy, then severely trampled; died right away, or declined from then on and died eventually
b	18	11	Trees where trampling contributed to death but was not necessarily the only cause: were unhealthy already, then trampled and died
11	5	3	Herbivory: Trees for which herbivory was the likely cause of death
12	8	5	Cage damage: Trees that were damaged by the cage, where this may have contributed to their death

Trees can fall into more than one category, and for this reason percentages do not add to 100. Trees in categories 1–6 were alive at the end of the study (30 September 2000); those in categories 7–12 died during the study period.

Shoot elongation showed a significant linear relationship with branch position on the tree, and therefore all values were adjusted for branch position before averages were calculated. To test whether the seasonal growth curves differed between blocks, treatments, and tree ages, profile analysis (a multivariate technique for analyzing repeated-measures data; von Ende 1993) was performed separately for each year. The total seasonal growth for each year was also analyzed with ANOVA, using the same methods and model as described above for damage scores.

Results

Germination And Survival Of Seeds

The first planting of seeds in 1997 resulted in the emergence of five seedlings from the 400 seeds planted, an emergence rate of 1.25%. None of the seedlings survived past the second week. Emergence rate was 17.5% for the second planting in 1997, with the seedlings evenly distributed over all blocks and treatments. All of these seedlings died within a few weeks except for one that survived until

midsummer 1998. No seedlings emerged from the ungerminated seed planted in April 1998, although several natural seedlings were observed in the vicinity of the plots.

Tree Survival

Mortality of planted trees in both age classes was greatest in the year of planting and decreased over time (Fig. 3). Overall, 64.4%, 44.4%, 39.4%, and 38.8% of trees survived the first, second, third, and fourth years, respectively. Life table analysis revealed no significant difference in survivorship curves between 4-year-old and 10-year-old trees (Wilcoxon chi-square = 0.1699, $df = 1$, $p = 0.680$). There were, however, significant differences among treatment groups (Wilcoxon chi-square = 22.507, $df = 7$, $p = 0.0021$). Survival was highest for trees planted 10 m away from the cliff edge in undisturbed locations (treatment 8); 95% of these trees survived the first year, and 85% were alive at the end of the study (Fig. 3a). Lowest survival (35% after year 1, 20% at the end of the study) was seen in trees planted at the cliff edge without supplements (treatment 1). All other treatments were intermediate, and only

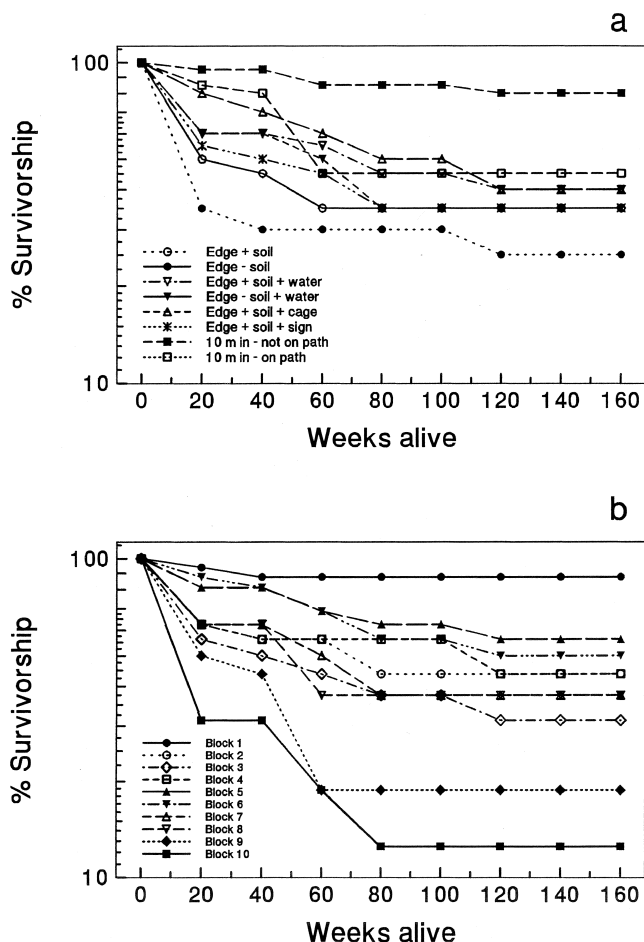


Figure 3. Survivorship curves for trees planted at the cliff edge in Bruce Peninsula National Park (a) by treatment and (b) by block. Means over both age classes are shown for each treatment or block.

the difference between the two extremes was significant ($p < 0.05$). Significant differences in survivorship curves were also found among blocks (Wilcoxon chi-square = 29.240, $df = 9$, $p = 0.0006$). Survival was highest in block 1, with 94% of trees surviving the first year and 87.5% the fourth year (Fig. 3b). Block 1 was significantly different from blocks 9 and 10, where 31% of trees survived year 1 and only 6.25% (corresponding to a single tree out of the 16 planted in this block) survived to the end of the study.

Tree Fate

Twenty-one percent of all trees became well established and were healthy throughout the study period (fate category 1; Table 2). Another 21% alternated between health and decline but were alive at the end of the study (fate 2). Fifteen percent of trees failed to become established (fate 7), and another 9% died without obvious external causes after alternating periods of health and decline (fate 8). Damage from dislodged or trampled cages probably contributed to the death of 5% of trees (fate 12). The remain-

ing trees died from external causes such as uprooting, trampling, or herbivory.

Thirty-seven percent of all trees were partially or completely uprooted at some point during the study period, and this was the most frequent external cause of tree death (fate 9). Sixteen percent were completely uprooted and either died immediately or disappeared (fate 9a and b), and in another 8% partial uprooting led to a slow decline and eventual death (fate 9c). Twelve percent of trees survived a partial uprooting (fate 3).

Twenty-five percent of all trees were severely or repeatedly trampled at some point during the study period. For 7%, this was almost certainly the reason for their subsequent death (fate 10a); 7% survived it (fate 4), and for 11% it was a contributing factor but not necessarily the only reason for their death (fate 10b). Seven percent of trees experienced serious herbivory at some point during the study period; herbivory was the most likely reason for the death of 3% of trees (fate 11). Four percent experienced similar levels of herbivory but survived (fate 5; Table 2).

The different fates were not homogeneously distributed over the eight treatments (Fig. 4), and tests of independence confirmed that some of these differences were significant (Table 3). Comparisons of the number of trees that survived and were healthy throughout (fate 1) versus the number of trees that died without external causes (fates 7 and 8) for different treatments show how the establishment and survival of trees are affected by the treatments in the absence of external reasons for mortality. Trees were more likely to be successful in the presence of added water, added soil, or both, whereas failure was more likely in the absence of these factors. Establishment success was significantly lower at the cliff edge than 10 m away from the edge; away from the edge, it did not differ between pathways and undisturbed locations (Table 3).

Herbivore damage, whether fatal or not, was rare at the cliff edge but much more common 10 m away from the cliff edge: 22.5% of all trees in treatments 7 and 8 experienced herbivory, as opposed to only 2.5% of all trees in the other six treatments. This difference was significant in a Fisher's exact test ($p = 0.0002$).

The frequencies of the different fates also differed across the study area. Blocks 1 and 2 as well as 5 and 6 had a low proportion of trees that failed to become established or died without external causes and a high proportion of trees that were healthy throughout, whereas the opposite was true for blocks 3 and 4 as well as 7–10. Blocks 8–10 had a high incidence of trees trampled or uprooted but a low incidence of herbivory (Fig. 4b).

There was also a significant relationship between establishment success (as measured by the relative frequencies of fate 1 vs. fates 7 and 8) and the characteristics of the planting site ($p < 0.0001$, Fisher's exact test). Trees planted in shady sites were much more likely to be healthy throughout and much less likely to die soon without external causes than trees planted in sites classified as open.

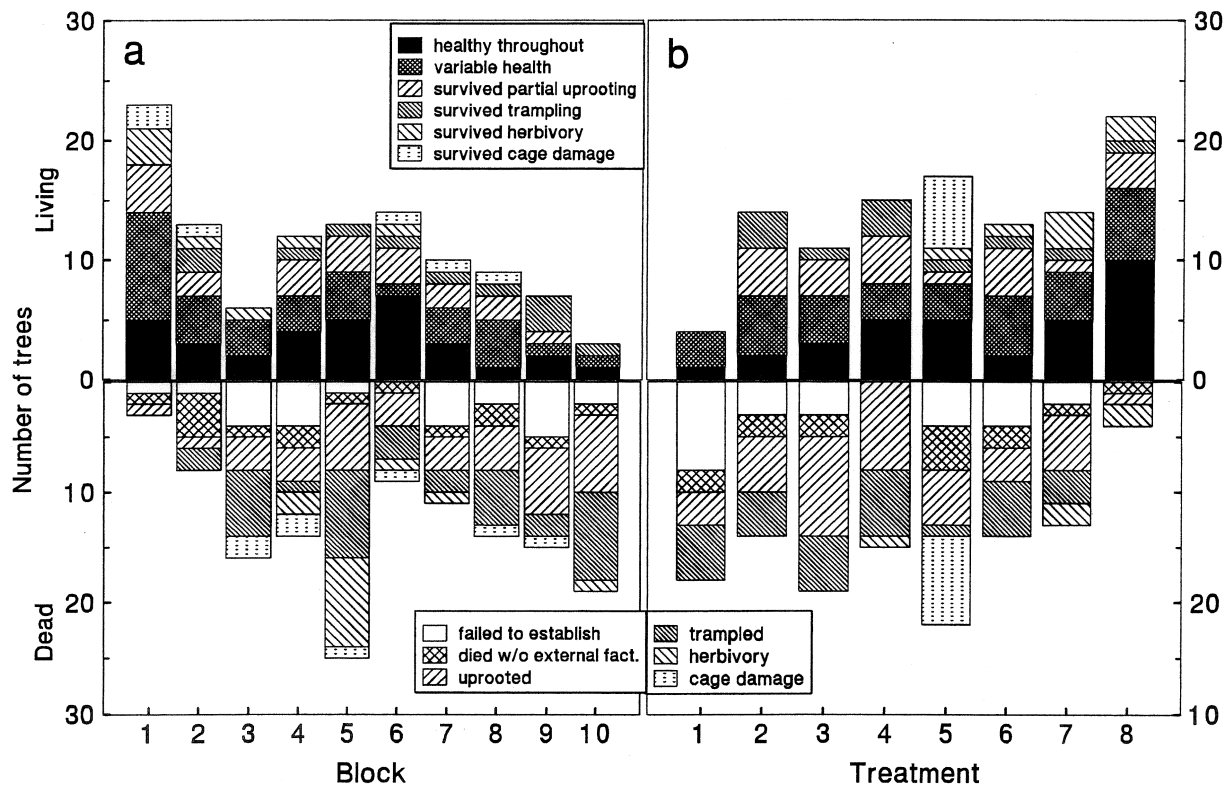


Figure 4. Number of trees in each block (a) or treatment (b) that fall into each of the 16 fate categories. Trees alive at the end of the study period are drawn above the horizontal axis; those that died during the study period are drawn below.

Damage to Living Trees

Damage to living trees by breakage increased over the first summer and leveled off after the end of September (Fig. 5). The pattern of damage was similar in the second and third years, but absolute ratings remained lower. Browning also increased steadily after the trees were planted and peaked in early autumn; in the second year,

initial scores were higher in the spring and the seasonal increase was smaller. Herbivore damage was highest in the year of planting and was greatest in the spring and lowest in midsummer. Root exposure ratings increased over the first summer and remained constant thereafter.

All types of damage showed the highest ratings and the greatest variation in the first year of the study (1997). Over subsequent years, the differences among blocks, treatments, and tree ages in the amount of damage received decreased. The most likely reason for this decrease is that the trees in the most exposed locations had died and were no longer included in the measurements. For this reason, we focus here on the results from the end of the first growing season (mid-September 1997), even though data were collected throughout the entire study period.

Breakage scores at the end of the first growing season showed significant differences due to tree age ($F = 4.23$, $df = 1, 101$, $p = 0.0668$) and age $\times$ treatment ($F = 2.36$, $df = 7, 101$, $p = 0.029$). Small trees experienced greater breakage than large trees in all treatments except the caged treatment, where the relationship was reversed. Caging significantly decreased breakage in small trees compared with the controls ($F = 9.04$, $df = 1, 101$, $p = 0.0033$ for contrast between treatments 5 and 2 in 4-year-old trees). In contrast, caging increased breakage in large trees, although not significantly ($F = 5.076$, $df = 1, 101$, $p = 0.0871$ for con-

Table 3. Results of Fisher's exact tests of independence between treatments and the likelihood of successful establishment and survival of planted trees, given the absence of external causes of mortality.

Effect Tested	Fates Compared	Treatments Compared	pValue (two-tailed)
Improved growth conditions	1 vs. 7, 8	1 vs. 2, 3, 4	0.047
Added water	1 vs. 7, 8	1, 3 vs. 2, 4	0.021
Added soil	1 vs. 7, 8	1, 2 vs. 3, 4	0.056
Cage	1 vs. 7, 8	2 vs. 5	1.000
Sign	1 vs. 7, 8	2 vs. 6	1.000
Distance from cliff edge	1 vs. 7, 8	2 vs. 7	0.041
Trampling	1 vs. 7, 8	7 vs. 8	0.262

Each test represents a 2×2 contingency table, where the number of trees in fate class 1 (healthy throughout) was compared against the number in the combined classes 7 and 8 (trees that failed to become established or died without external causes).

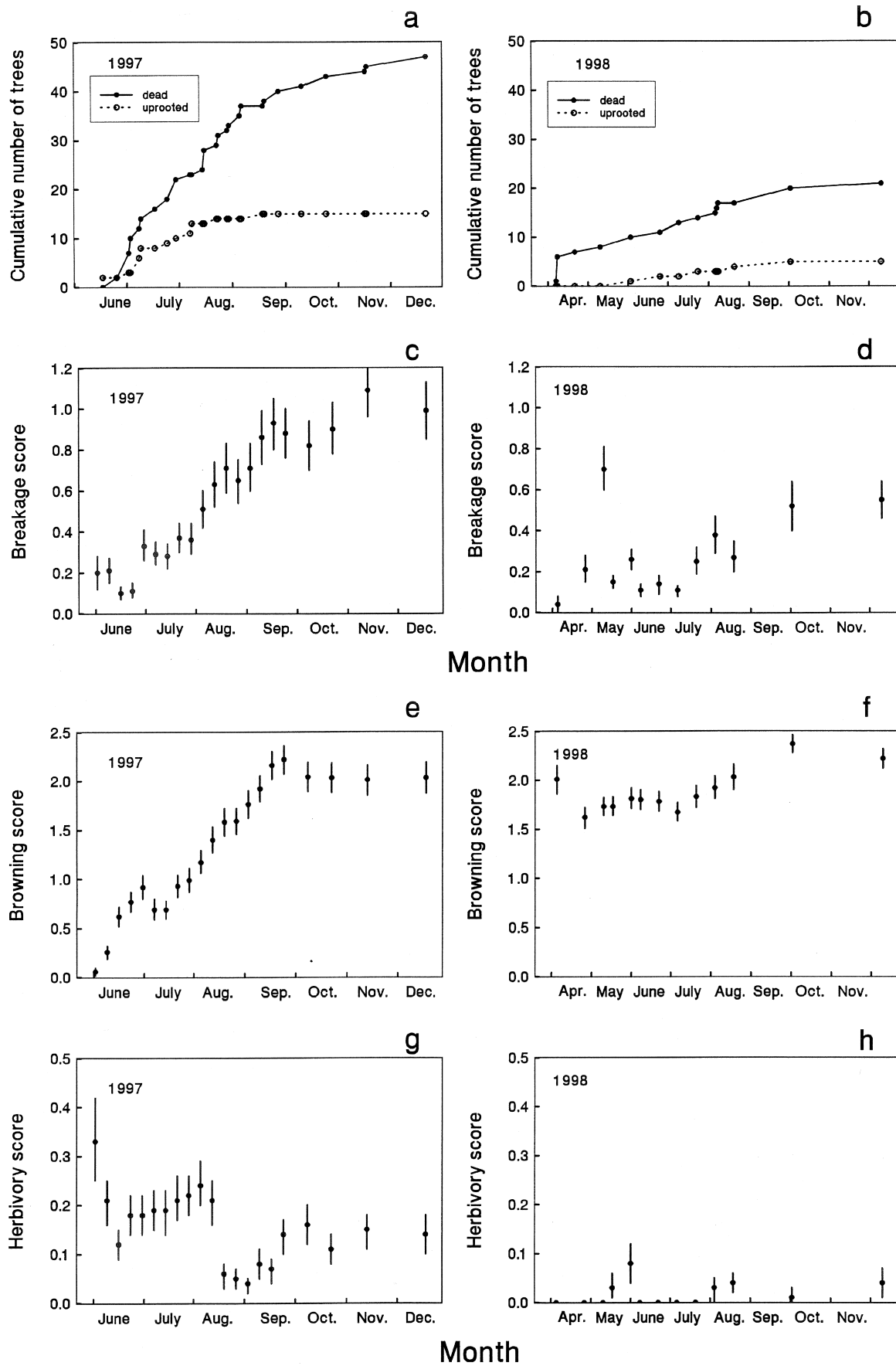


Figure 5. Seasonal patterns of (a and b) mortality by uprooting vs. other causes, (c and d) breakage scores, (e and f) browning scores, and (g and h) herbivory scores over the first two growing seasons. Values for c-h are means over all trees alive at each census date ($\pm$ SE).

trast between treatments 5 and 2 in 10-year-old trees). Breakage scores were low in block 1 and high in blocks 9 and 10, but this trend was not significant ($p = 0.098$).

Browning scores at the end of the first growing season were significantly higher in 10-year-old trees than in 4-year-old trees ($F = 12.78$, $df = 1, 93$, $p = 0.0006$). There was also a significant treatment effect on browning ($F = 3.94$, $df = 7, 93$, $p = 0.0008$) but no interaction between age and treatment. Contrasts showed a significant difference in browning between undisturbed and disturbed locations 10 m away from the cliff edge ($F = 5.47$, $df = 1, 93$, $p = 0.0214$ for treatments 7 vs. 8) but also between disturbed locations at the cliff edge versus 10 m away ($F = 7.97$, $df = 1, 93$, $p = 0.0058$ for treatments 7 vs. 2). Soil addition also significantly decreased browning scores at the end of the first growing season ($F = 5.09$, $df = 1, 93$, $p = 0.0264$ for comparison of treatments 1 and 3 vs. 2 and 4). Watering, caging, and signs had no significant effect (all $p > 0.05$). Browning was lowest in blocks 1 and 2 and highest in blocks 8–10, and the block effect was statistically significant ($F = 2.20$, $df = 9, 93$, $p = 0.029$).

The tendency of planted trees to fall over during their first growing season, as measured by the angle of the main axis with respect to the horizontal plane in autumn 1997, was much greater for large trees than for small trees ($F = 77.78$, $df = 1, 108$, $p < 0.0001$ for age effect). There were no significant differences due to block within the study area, but treatment significantly affected the angle of the main axis ($F = 2.56$, $df = 7, 108$, $p = 0.0177$). Trees planted without soil were the most likely and those planted 10 m away from the edge least likely to fall over, but the differences were not large enough for any of the contrasts to be significant. The angle of the tree at the end of the first growing season did not show any relationship with whether the tree lived or died during the study (chi-square = 2.188, $df = 1$, $p = 0.139$).

Root exposure ratings at the end of the first growing season were twice as high for 10-year-old as for 4-year-old trees ($F = 11.34$, $df = 1, 108$, $p = 0.0010$ for age effect in the ANOVA). Treatment and block effects were not significant, but the greatest root exposure was seen at the cliff edge, in the absence of supplements, and in blocks with high visitor density. There was also a trend for high root exposure in blocks 8–10 and low root exposure in blocks 1 and 2, but the block effect was nonsignificant as well.

Growth Rates of Surviving Trees

Shoot growth in surviving trees showed the typical seasonal course with fastest rates early in the year and a slowing down by midsummer (Fig. 6). Total seasonal growth showed no significant differences among treatments, blocks, and tree ages (overall p of the ANOVA, >0.10) for each of the 3 years that measurements were taken. The profile analysis addressed the question of differences in the shape of the seasonal growth curves among different blocks, tree ages, treatments, and treatment $\times$ age combinations by examining the interactions of each of these fac-

tors with time. All these interactions were nonsignificant ($p > 0.05$) in all 3 years, except for the age $\times$ time interaction in 1997 ($p = 0.015$). This indicates that block and treatment did not affect seasonal growth patterns of surviving trees but that small and large trees differed in their growth curve in the year of planting. Contrasts for the individual time intervals showed that the age $\times$ time interaction was only significant for the very first time interval in 1997, the month immediately after planting. Growth rate during this period was significantly faster in 4-year-old trees than in 10-year-old trees; thereafter, growth rates were statistically the same for both age groups. For the following 2 years, there were no significant differences in growth rates among age groups, but 10-year-old trees tended to grow slightly faster (Fig. 6).

Visitor Counts

The total number of visitors counted in the 10 blocks over the four observation periods ranged from 5 to 789 (Fig. 2). Block 10, at the Grotto, received the highest number of visitors, followed by block 6 where the Bruce Trail first accesses the shoreline. The lowest visitor numbers were found in blocks 1–3 (Fig. 2).

Discussion

Effects of Visitor Density

The results show that a low survival rate of *T. occidentalis* is correlated with high visitor numbers in the more popular sections of cliff edge in Bruce Peninsula National Park. The visitor counts showed a more than 150-fold range in visitor numbers as a result of the visitor flow patterns within the study area. The corresponding range in restoration success was approximately 15-fold, from a 4-year survival rate of 87.5% in block 1 to 6.25% in blocks 9 and 10. The large effect of disturbance on restoration potential is underlined by the result that for almost all measures of restoration success, differences among blocks within the study area exceeded all differences among treatments.

A study of visitor distribution in Bruce Peninsula National Park that used both surveys and trail counter data (Kettle 1998) supports our own results that the Grotto (block 10) was the most popular location along the Georgian Bay shoreline. Most visitors access the cliff edge at block 6 and then turn west to pass through blocks 7 and 8 toward the Grotto; blocks 8 and 9 are often bypassed because the shoreline rises and forms a point there. Visitors may also walk eastward along the cliff edge from the access point in block 6, but their numbers decline progressively toward block 1. These patterns of visitor density explain much of the variation in tree survival, fate, and damage seen within the study area. Block 1, with the lowest visitor density, had the highest 4-year survival rate, the fewest trees that failed to become established and the lowest scores for browning, breakage and root exposure. Block 10 had the lowest survival rate and the highest dam-

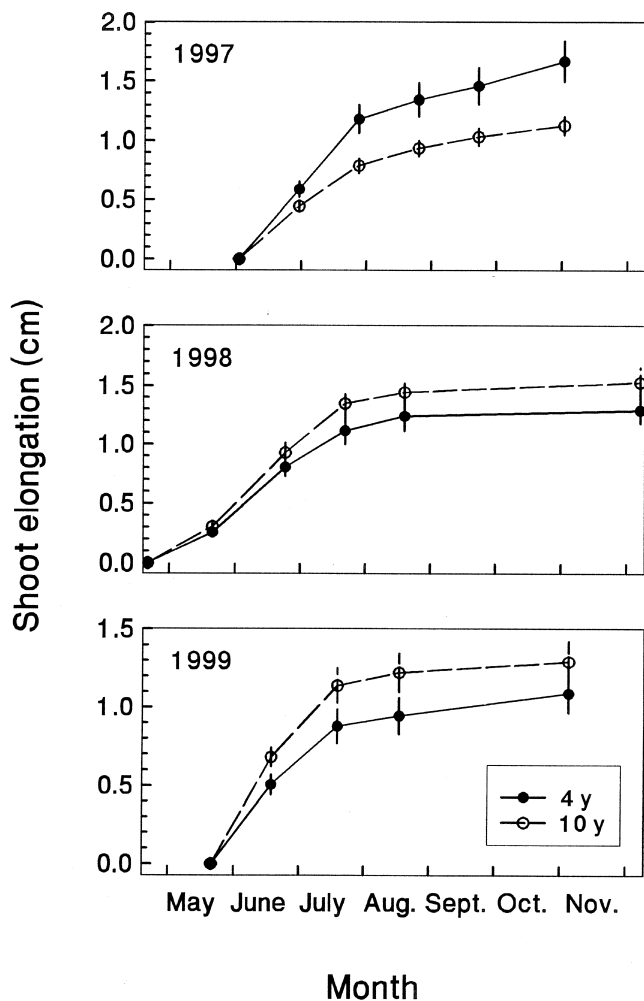


Figure 6. Patterns of seasonal shoot elongation in 4-year-old (●) and 10-year-old (○) planted trees for the first 3 years after planting. Elongations shown are means ($\pm$ SE) over all treatments and blocks and are cumulative over each growing season.

age scores. However, although visitor impact was clearly the major factor causing mortality of planted trees, it was not the only factor, and not all results are explained by visitor numbers. For example, survival rates in block 9 were as low as in block 10, even though block 9 had much lower visitor numbers. Conversely, block 6 had a low frequency of trees that failed to become established and a high frequency of trees that were healthy throughout, even though it received the second highest number of visitors. Visitor residence time may play a role in these discrepancies, as visitors lingering in an area are likely to have a greater impact than those passing through quickly, as most do in block 6.

Effects of Planting Site

The planting site was the second most important factor affecting the survival, fate, and damage scores of planted trees. In this experiment, two aspects of the planting site

were controlled by the treatment: the distance from pathways and the distance from the cliff edge. A third aspect, the presence of shade from nearby canopy trees, was not controlled but was recorded for each planted tree.

The effect of trampling was only studied away from the cliff edge because no undisturbed sites were available at the cliff edge. Planting sites on pathways were less favorable for the successful establishment of *T. occidentalis* than were untrampled planting sites. Besides ongoing disturbance, this may be due to decreased soil depth and litter cover and increased soil bulk density (Parikesit et al. 1995). Soils compacted by trampling are generally considered unfavorable for seedling establishment and early growth (Tuttle et al. 1988); on the other hand, *T. occidentalis* is relatively resistant to trampling, and its establishment may actually be favored over that of other species by soil compaction (de Blois & Bouchard 1995). The same is true for *Juniperus virginiana* (Schmidt & Stubbendieck 1993).

Trees planted at the cliff edge were more likely to die soon after planting or suffer root exposure and poor health and had lower 4-year survival rates than trees planted in equally disturbed locations 10 m away from the cliff edge. A number of factors may be responsible for this pattern, which is also seen in natural seedling establishment at cliff edges (Bartlett et al. 1991). Shallow soils near the cliff edge may exacerbate root exposure due to trampling, which in turn makes trees more susceptible to drought and wind throw (James et al. 1979). Microclimatic conditions become more severe at the cliff edge, in particular water availability, temperature fluctuations, and wind speed (Bartlett et al. 1990). In Bruce Peninsula National Park, this natural gradient may have been further increased by the absence of canopy-forming trees from the most disturbed cliff edges. The analysis showed that trees planted in shady conditions, regardless of treatment or block, were more likely to remain healthy throughout the study period than trees planted in open areas. In blocks 5 and 6, where survival rates were better than expected from the high visitor density, more than half the trees had been planted in the shade; conversely, all trees in block 9, which had lower than expected survival rates, had been planted in open locations. All of the trees in the fate category "failed to become established" had been planted in open conditions and none in the shade. Although a tree canopy is generally known to create conditions favorable for tree establishment (McChesney et al. 1995), such a large effect was not expected for *T. occidentalis*, which is said to regenerate best on exposed mineral soils with abundant light (Johnston 1990). The only advantage of the cliff edge as a planting site was the absence of herbivory by rabbits or rodents; however, herbivory was not a major cause of mortality in this study.

Plant Material for Restoration

For large-scale restorations, seeds are often preferred because of the low cost and effort required in procuring,

transporting, and planting them (Harker et al. 1993). However, the success of this method was extremely low in the present study. Very few *T. occidentalis* emerged from seed, regardless of treatment, disturbance, and whether or not seeds were pregerminated, and none of them survived for more than a year. Walmsley and Davy (1997a) also concluded that seeding directly in the field could not be recommended for restoration of a coastal shingle beach, especially for perennial species, while planting of container-grown seedlings was effective. Trees require a long time to become established from seed, and for this reason alone seeds may be a poor choice for restoring habitats in the presence of continued disturbance. If disturbance is discontinued, forested habitats can be restored successfully by broadcasting tree seeds (Camacho-Cruz et al. 2000). Whether or not restoration from seed is a realistic possibility also depends on the species: Camacho-Cruz et al. (2000) found that success rates were different for different species, and *T. occidentalis* has very low recruitment rates in nature (Matthes, unpublished data). For the container-grown *T. occidentalis*, tree age made no difference to overall success rate. Younger trees are the material of choice because they are less expensive to plant and become established more quickly once planted.

Efforts to Improve Growth Conditions

Soil surface treatments are often recommended to increase seed retention and seedling establishment in restoration projects (Chambers & MacMahon 1994). Although soil addition or watering did not make seeding a viable restoration technique, it did slightly improve first-year survival and fate of container-grown trees; trees planted without supplements of any kind did the worst in all respects, whereas trees receiving both water and soil did best. Browning of the trees was decreased by soil addition but not by supplemental watering. Soil addition also decreased the tendency of planted trees to fall over. However, fallen-over trees had no higher rates of mortality than trees growing upright; this may be because *T. occidentalis* commonly grows horizontally or even in inverted direction on cliff faces (Kelly et al. 1992).

Unlike survival, the growth rate of planted trees was not affected by either soil or water and in fact showed no differences among any of the treatments or planting locations in this study. The result that supplemental resources may affect survival but do little to increase growth rate is in agreement with previous work on *T. occidentalis* (Matthes-Sears et al. 1995; Matthes-Sears & Larson 1999).

Efforts to Protect Planted Trees

Getting visitors to refrain from damage-causing activities is a common challenge on public land with high visitor traffic (Mortensen 1989), and efforts to protect plantings from human damage were of only limited success in this experiment. Regardless of treatment group, many planted

trees were actively and in some cases violently dislodged or uprooted by visitors. Visitors appeared unresponsive to the signs conveying appeals not to disturb the trees: Signs had no effect at all on tree survival, fate, damage, or growth. The high frequency with which the signs were trampled or dislodged suggests they may need to be larger to be noticed.

Physical protection of trees by caging had mixed success. The open-topped enclosures did not protect the trees from herbivores, but herbivory was not a major cause of tree loss at this site. Cages were frequently found at the bottom of the cliff or suspended in the canopy of neighboring trees after having apparently been removed by visitors. For the smaller trees, cages significantly decreased the amount of breakage, most likely because visitors were encouraged to walk around rather than step over the trees. However, the survival rate of these trees was unaffected by caging, as were fate, browning, and growth rate. For large trees the enclosures were detrimental, increasing the amount of breakage compared with uncaged trees. Most likely, this was because these trees were too tall to be stepped over even when uncaged, and the taller cages were more likely than shorter ones to be removed by visitors or blown over by the wind, causing breakage to the trees within them.

Management Recommendations

We suggest the planting of smaller (e.g., 4-year-old) container-grown trees without supportive measures of any kind for cliff-edge restoration in Bruce Peninsula National Park. Although water and soil addition made a small improvement in survival, this improvement was disproportionately small considering the effort involved in carrying water or soil to the planting sites. The same was true for protective cages, whose small improvement to survival was far outweighed by the effort required for setup and maintenance in addition to the aesthetic drawback. The size and age of container-grown trees are unimportant, but smaller trees are recommended because they are easier to grow, transport, and plant. The broader implication of these results is that, in other settings, the age-related patterns of mortality should be carefully examined in the context of the site conditions *before* the restoration materials and protocols are selected.

The large differences in tree survival between trampled and untrampled planting sites suggest that, in conjunction with reforestation, the current network of small pathways should be blocked off using boulders or brush and replaced by a single clearly marked main trail or boardwalk. The effectiveness of such visitor channeling is apparent from the high planting success in the heavily visited blocks 5 and 6, where a dense forest in combination with topography has so far largely prevented the networking of pathways so evident in other parts of the study area. Since shade from existing canopy trees was shown to favor restoration efforts, one management priority should be to pre-

vent the further degradation of cliff edges to the point where few or no canopy trees remain. Where such trees are absent, as they are currently in the most disturbed parts of the study area, restoration may be difficult under current circumstances and a temporary reduction in visitor admissions may need to be considered. Such a reduction would be most beneficial during the summer holiday season and into the autumn, when our data show visitor impacts to be greatest.

A drastic reduction of visitor admissions to Bruce Peninsula National Park is only one possible way to improve the success of a future cliff-edge restoration project. The fate data indicate that a large proportion of tree mortality results from careless (trampling of trees) or even intentionally destructive (uprooting of trees) visitor behavior. If park managers find a way to specifically target these behaviors, for example, through increased efforts at visitor education in combination with increased surveillance at peak times, it may be possible to have successful site restoration coexist with large visitor numbers enjoying the park in appropriate ways.

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